



2 INTELLIGENT DRONE For Industrial Inspections

An industrial setup requires strict vigilance of the functioning of all operational units. A single undetected problem in the setup can lead to long-lasting issues like long operational downtime, hassles with repair and maintenance of machines and, in critical situations, health and life risks to workers. Repercussions of such setbacks often lead to heavy financial losses, delay in deliveries, compromised quality, hassle to the entire workforce and a situation of complete chaos.

In functional areas like oil and gas factories, manufacturing facilities, warehouses, aerospace industry, and especially in mission-critical industries like nuclear plants, defence or Big Data centres, the slightest malfunctioning of equipment can lead to any of the earlier mentioned difficulties or, worse, all of them together. While smart solutions and Industrial Internet of things (IIoT) platforms have given factory runners the capability to monitor the assets in real time, deployment of these solutions is often time-consuming and costly. Moreover, deploying such solutions is often difficult in remote areas or on the field. To cover up these gaps and provide accurate proactive reports in real time, Chennai-based DeTect Technologies have created an intelligent drone called Noctua.

The startup, with the handholding of IIT-Madras, was able to create



the smart drone for industrial needs. Noctua homes a small camera that can capture and generate high-quality images, which can be analysed for reports. Sensors in the drone can detect faults like leakages, malfunctioning machines, loose connections and more.

The camera also comes with thermal vision to accurately detect the source of problems in the setup. It can be used to inspect such elements like pipelines, vessels, columns, reactors, operation lines and the like. Flight paths are auto-memorised so that the drone can cover all necessary areas of the factory floor or on the field during inspection.

AI-powered algorithm programs the drone to traverse these routes and capture necessary data during inspection. The algorithm also provides high

sensitivity towards nearby obstacles, enabling collision avoidance with precision. This makes Noctua suitable for indoor flight and inspection.

Acquired data is stored in the cloud in real time. Analytics of the data are also generated on the cloud and can be accessed by users to get reports of their system performance. Computer vision program allows it to analyse and detect faults via image analysis, and raise alerts immediately in emergency situations. Noctua platform enables factory runners to conduct inspections in high temperature areas, remote locations and blind corners of the factory. Customisable for various industrial use-cases, Noctua can make large-scale setups error-free and smooth.

—Paromik Chakraborty, technical journalist, EFY

3 INNOVATIVE AIR PURIFIERS To Improve Indoor And Outdoor Air Quality

Air quality index (AQI) is a measure of air quality used by governments and regulatory bodies to indicate pollution levels in

a particular area. The higher the AQI, the less fit the air quality to live in that particular area. Poor air quality affects public health and increases the

chances of airborne diseases.

In general, AQI is a composite of air pollutants like ozone, carbon-monoxide, sulphur-dioxide, nitrogen-